An exact endpoint diagnostic for the Toda momentum and phase dictionary

Quantum Boussinesq solitons project

4 November 2026

Abstract

The implemented Toda Fredholm equations imply an exact edge momentum and an endpoint reciprocity identity for their directional phases. **(c)** These fix the symmetric part of a possible quantum frame correction. They do not fix its antisymmetric part or identify it with a physical charge transformation. **(d)** An independent 60-digit calculation finds a nonseparable symmetric remainder after the classical frame subtraction actually used in the overtaking code. **(c)** This shows that the retained classical subtraction does not antisymmetrise the directional kernel at finite $\hbar$; it does not by itself refute the forward ordered physical amplitude or determine the edge limit. The existing model, momentum prescription, and normalizations are retained; no replacement physical convention is selected. **(d)**

1 Definitions and scope

All results below are consequences of the existing asymptotic Bethe-ansatz integral equations, not a new proof of their physical applicability. **(c)** denotes project derivations and checks; **(d)** marks unresolved physical identifications. The core high-precision checker `audit_toda_dictionary.py` is independent; an isolated final comparison uses the original implementation. It records input hashes in `audit_toda_dictionary_results.json`. No external literature claim is introduced.

Use $h = \hbar_T$, $t = 2\pi h$, and the quantum ground-state interval $[-q, q]$. The code `toda_edge_ba.py:26--67` defines

$$\Theta(d) = 2h\ \mathrm{Im}\log\Gamma(1 + id/h) + 2d\log h, \qquad K(d) = \Theta'(d), \tag{1}$$

$$(\mathcal{A}f)(x) = tf(x) - \int_{-q}^{q} K(x-y)f(y)\,\mathrm{d}y, \qquad \mathcal{A}\rho = 1, \qquad \int_{-q}^{q}\rho = 1/a, \tag{2}$$

$$\mathcal{A}F_p(x) = \Theta(p-x), \qquad P(p) = p + \int_{-q}^{q}\Theta(p-x)\rho(x)\,\mathrm{d}x, \tag{3}$$

$$\Phi(p,z) = \Theta(p-z) - \int_{-q}^{q} K(p-x)F_z(x)\,\mathrm{d}x. \tag{4}$$

Here p is the bare Bethe spectral parameter. $P(p)$ is its dressed momentum. Assume the ground-state integral operator is invertible and its solutions are smooth up to the finite endpoints, as in the tested case. The even kernel makes $\mathcal{A}$ self-adjoint for the unweighted integral inner product. The ground density is even, and P is odd.

2 The edge subtraction in momentum is exact

Differentiation and the ground-state equation give, inside the sea, $P'(p) = t\rho(p)$. Integrating this identity from $-q$ to q and using oddness yields **(c)** (`audit_toda_dictionary.py`)

$$\boxed{P(q) = \frac{\pi h}{a}, \qquad P(-q) = -\frac{\pi h}{a}.} \tag{5}$$

Self-adjointness also gives $P(p) = p + \int_{-q}^{q} F_p(x)\,\mathrm{d}x$, the total momentum including backflow. Thus the relative physical momentum used by the code is a subtraction of the known quantum Fermi momentum, rather than a new definition of the bare coordinate. **(c)**

For a right-moving excitation, the implemented label is fixed through

$$P(p) - P(q) = \frac{4b}{a}(\kappa\cosh\kappa - \sinh\kappa), \quad b = e^{-a/2}, \quad \kappa = \sqrt{yh/b}, \quad u = \frac{p-q}{h}. \tag{6}$$

These are the existing $\kappa = \sqrt{3}\beta$, $h = 3b\hbar_B$, $y = \beta^2/\hbar_B$ conventions. Differentiating at fixed ground state gives an exact Jacobian **(c)** (`audit_toda_dictionary.py`)

$$\frac{\mathrm{d}u}{\mathrm{d}y} = \frac{2\sinh\kappa}{aP'(p)}. \tag{7}$$

This specifies what an asymptotic derivation of $u(y)$ must invert; identifying u with dressed physical momentum would skip this Jacobian. Equation (5) alone neither proves nor disproves the proposed large-y logarithmic expansion for $u(y)$. **(d)**

There is a small implementation distinction: `edge()` returns $e = q(1 + 10^{-10})$, not q. Its displacement in edge units is exactly $(e - q)/h = 10^{-10}q/h$. **(c)** The offset is tiny in the saved runs but is not uniform in a literal $h \to 0$ limit. A controlled edge derivation must specify that numerical offset tends to zero faster than h, or use the exact endpoint analytically. This audit changes no production code. **(d)**

3 Exact reciprocity and its proof

For arbitrary real external arguments define the symmetric dressed kernel

$$\mathcal{R}(p,z) = K(p-z) + \int_{-q}^{q} K(p-x)\,[\mathcal{A}^{-1}K(z-\cdot)](x)\,\mathrm{d}x.$$

Differentiate (4), and integrate once by parts in the equation for F_z. The resulting identities are **(c)** (`audit_toda_dictionary.py` checks the integrated consequences below)

$$\partial_z\Phi(p,z) = -\mathcal{R}(p,z), \tag{8}$$

$$(\partial_p + \partial_z)\Phi(p,z) = \mathcal{R}(p,q)F_z(q) - \mathcal{R}(p,-q)F_z(-q), \tag{9}$$

$$\partial_p F_p(\pm q) = \mathcal{R}(\pm q, p)/t. \tag{10}$$

Consequently the derivative in either argument of $\Phi(p,z) + \Phi(z,p)$ equals that of the following endpoint bilinear. The integration constant vanishes at $p = z = 0$ by parity. Hence **(c)**

$$\boxed{\Phi(p,z) + \Phi(z,p) = t\{F_p(q)F_z(q) - F_p(-q)F_z(-q)\}.} \tag{11}$$

This retains the finite-sea boundary terms. Discarding them would incorrectly make the directional Bethe phase antisymmetric.

Let e_1, e_2 be either exact edge, and set

$$\widehat{\Phi}(p,z) = \Phi(p,z) - \Phi(p,e_2) - \Phi(e_1,z) + \Phi(e_1,e_2), \tag{12}$$

$$f_p^{\pm} = F_p(\pm q) - F_{e_1}(\pm q), \qquad g_z^{\pm} = F_z(\pm q) - F_{e_2}(\pm q). \tag{13}$$

The reverse phase uses exchanged references. Subtracting four copies of (11) gives **(c)** (same-side and opposite-side checks)

$$\widehat{\Phi}(p,z) + \widehat{\Phi}(z,p) = t(f_p^+ g_z^+ - f_p^- g_z^-). \tag{14}$$

For clarity, write this as $\mathfrak{M}_1\mathfrak{Q}_2 + \mathfrak{M}_2\mathfrak{Q}_1$ using the computable response functions

$$\mathfrak{M}(p;e) = ta\rho(q)\{F_p(q) - F_p(-q) - F_e(q) + F_e(-q)\}, \tag{15}$$

$$\mathfrak{Q}(p;e) = \frac{F_p(q) + F_p(-q) - F_e(q) - F_e(-q)}{2a\rho(q)}. \tag{16}$$

Integration of the differentiated F equation gives $t\rho(q)[F_p(q)-F_p(-q)] = 1-P'(p)$, so equivalently **(c)**

$$\mathfrak{M}(p;e) = -a[P'(p) - P'(e)]. \tag{17}$$

These symbols denote spectral response functions, not newly established physical compression and momentum charges. **(d)** Their normalization is useful for comparison: the classical expressions have $P_0'(p) = 2\kappa/a$, $P_0'(2b) = 0$, and $-a\partial_a P_0(p)|_p = P_0(p) - 2b\sinh\kappa$, verified symbolically in `audit_toda_dictionary.py`. **(c)** This does not prove a uniform limit of the quantum endpoint functions.

Any frame correction intended to make the four-term phase antisymmetric must have the symmetric part (14). The product $\mathfrak{M}_1\mathfrak{Q}_2$ is an algebraic example with that symmetric part. Adding an arbitrary antisymmetric two-body function leaves the condition unchanged. No such product is selected here as the physical quantum frame, and the identity supplies no Ward identity relating these functions to continuum quantum charges. **(d)**

4 The actual two-term overtaking subtraction fails this test

The overtaking code `toda_edge_twoterm.py:11--17` uses only $N(p,z) = \Phi(p,z) - \Phi(p,e_2)$. For its reverse directional expression, an exact rearrangement gives **(c)** (`audit_toda_dictionary.py`)

$$N(p,z) + N(z,p) = \widehat{\Phi}(p,z) + \widehat{\Phi}(z,p) + U(z) + V(p), \tag{18}$$

$$U(z) = \Phi(e_1,z) - \Phi(e_1,e_2), \qquad V(p) = \Phi(e_2,p) - \Phi(e_2,e_1). \tag{19}$$

Thus two-term and four-term normalizations have identical mixed dependence; they are not the same phase. A rectangular mixed difference removes U, V and any other one-body reference without selecting a convention.

At each fixed dressed momentum, retain the original classical frame

$$C_{\mathrm{cl}}(\kappa_1,\kappa_2) = (-2\kappa_1)\left[\frac{P_n(\kappa_2)}{a} - 2b\sinh\kappa_2\right].$$

Let $S_{\mathrm{rem}} = N(p_1,p_2)+N(p_2,p_1)-C_{\mathrm{cl}}(1,2)-C_{\mathrm{cl}}(2,1)$, where the bare p_j are obtained by inverting (6). For $\kappa_1 \in \{0.8, 1.0\}$ and $\kappa_2 \in \{0.35, 0.5\}$, define the mixed rectangle as the upper–upper minus lower–upper minus upper–lower plus lower–lower values. At $h = 0.4$, $q = 1.5$, the count equation fixes $a = 1.160312923180467\ldots$. Independent calculations give **(c)**

Gauss nodes	Largest endpoint identity error	Rectangle of S_{rem}/h
40	$1.93\,10^{-10}$	-0.1125712665447065
64	$4.42\,10^{-16}$	-0.1125712665272591
88	$1.09\,10^{-21}$	-0.1125712665272590

The original `EdgeBA` implementation at 120 nodes, including its fixed edge offset, gives $-0.1125712665368\ldots$, differing by less than 10^{-11}. The same independent rectangle is obtained directly from $\mathfrak{M}_1\mathfrak{Q}_2 + \mathfrak{M}_2\mathfrak{Q}_1 - C_{\mathrm{cl}}(1,2) - C_{\mathrm{cl}}(2,1)$. At the fixed spectral-to-dressed-momentum dictionary used here, it cannot be removed by one-body phase/reference subtractions, including switching between the displayed two- and four-term phase normalizations. **(c)** Changing the

momentum reference and re-inverting the bare parameters is a different operation and is not tested by this rectangle. The reverse directional phase here is used as a reciprocity diagnostic; it is not asserted to be a second overtaking collision in the same ordering sector. This test is necessary for a scalar antisymmetric frame identification, not a complete derivation of that identification. **(d)**

Any extension to a common antisymmetric scalar frame therefore requires an additional mixed symmetric correction at finite h. It does not determine whether its normalized correction vanishes, remains finite, or diverges in the particular nested edge and large-Y limits. The direct Toda expansion should use the implemented quantity as defined; comparison to continuum physical scattering still requires the antisymmetric frame part and the physical charge dictionary. **(d)**

5 Verification and failures recorded

`audit_toda_dictionary.py` uses exact SymPy identities and a fresh 60-digit mpmath Nyström solver. Fixing (h, q) and determining a from the count condition selects the same ground-state family as fixing (h, a) and solving for q. The dense quadrature, rather than decimal arithmetic, limits the displayed accuracy. Checks include backflow momentum, edge momentum, reciprocity, both reference orientations, the response derivative, physical-momentum inversion, and the actual classical-frame mixed rectangle. Intermediate resolvent and endpoint derivative identities are also checked directly. All pass.

Claim	Label	Verification
Dressed edge momentum and backflow	(c)	Independent Fredholm solve
Momentum Jacobian and classical responses	(c)	Exact symbolic derivatives
Reciprocity and endpoint bilinear	(c)	Mesh convergence and direct derivative checks
Two-/four-term phase relation	(c)	Exact rearrangement and numerical comparison
Classical-frame mixed remainder	(c)	Three meshes and original-code comparison
Physical frame and nested edge limit	(d)	Not established by these identities

Failures recorded:

1. Exploratory 24- and 40-node meshes gave reciprocity discrepancies about $3.5\,10^{-7}$ and $6.3\,10^{-11}$. Mesh refinement was required; the identity and subtraction were not changed to suppress those errors.

2. The implemented classical frame fails the nonseparable symmetric two-body test above. This failure is retained as the substantive result.

3. An initial PDF layout edit mismatched an equation environment. The LaTeX build stopped; the environment was corrected before the final build.

4. No implementation assertion failed, and no test tolerance was weakened. The fixed relative edge offset is recorded explicitly rather than treated as an exact endpoint in a formal limit.

The script ends with `ALL CHECKS PASSED` for the stated identities and detected discrepancy. Physical interpretation of the remaining antisymmetric frame and of the continuum quantum charges remains open. **(d)**